



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
Mid-Spring Semester Examination Re Test 2024-25

Date of Examination: _____ Session: (FN/AN) _____ Duration: 2 hrs.
Full Marks : 30 Subject No.: NA20202 Subject : Marine Hydrodynamics
Department/Center/School: Ocean Engineering and Naval Architecture
Specific charts, graph paper, log book etc., required
Special Instructions (if any): NO

Please note that the notations and variables used in the question paper are same as that of used in the class and hence not explained here.

1. Derive the Continuity equation for cylindrical co-ordinate system. If you consider the velocity potential in cylindrical system for 2-D case as $V_r = \frac{\partial \phi}{\partial r}$ and $V_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta}$. Derive the expression for Laplace equation. 5
2. Show that, at which condition, the velocity potential $\phi = rf(\theta)$ would be the possible form of velocity potential. 3
3. Suppose a stream of flow coming from $-\infty$ with a velocity $4m/s$ and it passes through a cylinder of radius 6m. Find out what would be the volume flow rate of the dipole to replicate the same phenomena mathematically. Also calculate the fluid pressure at the point (0,6) 4
4. A uniform flow is coming from $-\infty$ with a velocity $10m/s$. A source is placed at (0,0) and at (2,0) with volume flow rate $\pi m^2/s$. Find particle velocity at (1,1). 4
5. Show that $G(x,y,z) = \frac{1}{r}$ where $r^2 = x^2 + y^2 + z^2$, satisfies the Laplace equation everywhere except the origin and from that results, show that velocity potential $\phi(P)$ at any point P in an unbounded fluid can be given as $\phi(P) = -\frac{1}{4\pi} \left(\iint_S \left(\phi(Q) \frac{\partial}{\partial n} \left(\frac{1}{r} \right) - \frac{1}{r} \frac{\partial \phi(Q)}{\partial n} \right) dS \right)$. 6
6. Show that, if a body is moving with a velocity $\bar{V}(t)$, then the force experienced in the direction of i due to the movement of the body in the direction of j can be written as : $F_i = -\dot{V}_j m_{ij}$, also prove that $m_{ij} = m_{ji}$ 8